Diet and serum cholesterol: do zero correlations negate the relationship?

The confounding that results from the uncontrolled conditions under which most epidemiologic observations are made is sufficient to undermine their validity with respect to investigation of the relationship between diet and serum cholesterol. In this paper, the authors show, using both a mathematical model and referring to empirical data, that if certain variances are sufficiently great, even when there is cause and effect, correlation coefficients close to zero would be expected from the actual data of a cross-sectional study. Cross-sectional designs are therefore not suitable for studying this relationship.